COAXOLOTL

Mast-head LNA + 8-way filter bank, powered and keyed over one coax by the RTL-SDR bias-T.

Sheets (review order):
  2 Power - VCOAX tap, LM66100 ideal diode, TPS2553 + INA180 LNA supply, hold-up, 3V3 [this review]
  3 MCU   - EFM8BB21F16G, C2, reset-safe pulls, test points [this review]
  4 RF   - antenna protection, LNA/bypass SPDTs, preselector, FM notch, 2x PE42582, 8 filters, bias-T [next]
  5 Panel - 6x 74HC595, 48 lamps [next]

Conventions: nets in CAPS with global labels cross sheets. Every pin is labelled (no long wires),
so the Python in hardware/kicad/tools/build.py is the source of truth.
Firmware pin map: firmware/src/hal_efm8bb2.c Design notes: docs/hardware.md

Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

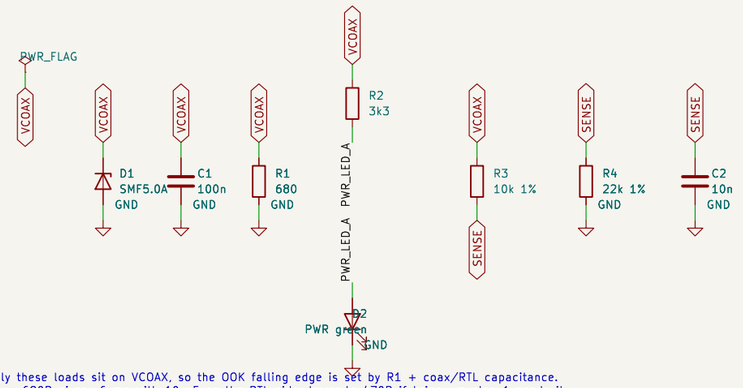
Generated by hardware/kicad/tools/build.py - edit the Python, not this file
Bias-T powered, OOK switched LNA + 8-way filter bank
z2labs

Sheet: /
File: coaxolotl.kicad_sch

Title: Coaxolotl

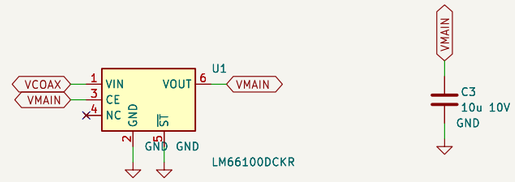
Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/3

A VCOAX tap (ahead of all bulk capacitance)



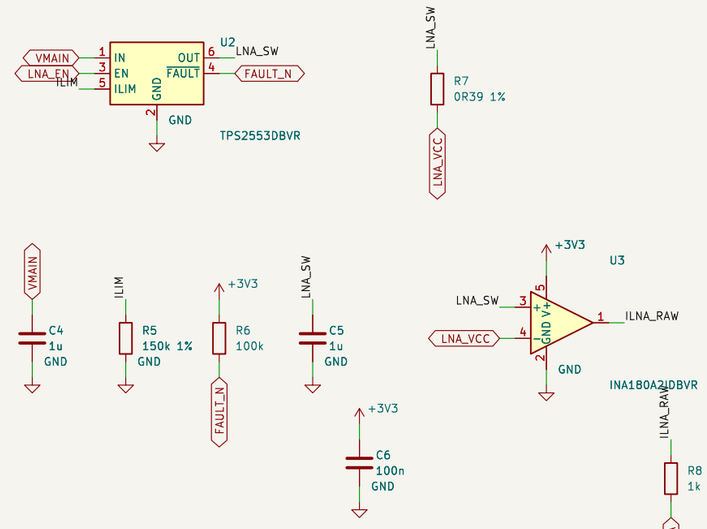
Only these loads sit on VCOAX, so the OOK falling edge is set by R1 + coax/RTL capacitance.
R1 = 680R gives ~6 ms with 10 uF on the RTL side; lower to 470R if bring-up step 1 needs it.
SENSE = 0.69 x VCOAX (3.0 V @ 4.4 V). Goes to P0.0 (port match) and P0.1 (ADC).

B Ideal diode VCOAX -> VMAIN



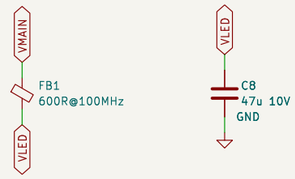
CE tied to VOUT: the comparator turns the FET off whenever VMAIN > VCOAX,
so LNA, lamps and hold-up never back-feed the sense node during a gap.
ST unused -> GND (datasheet).

C LNA supply: current-limited switch + shunt monitor



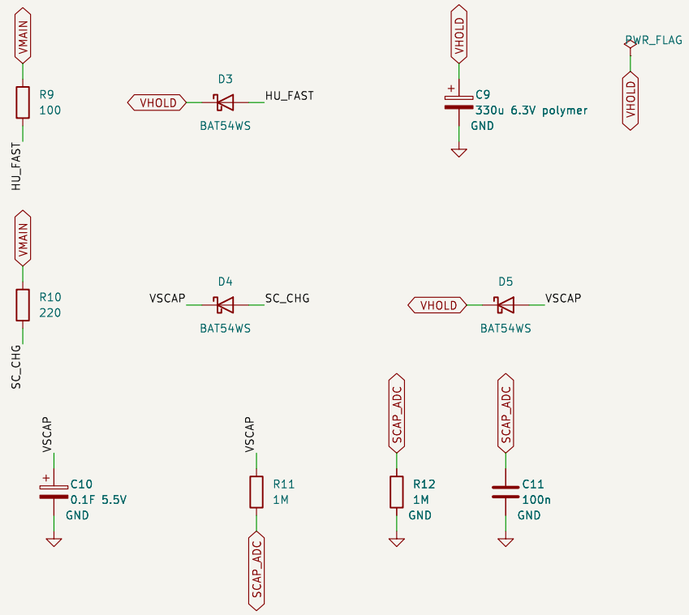
TPS2553: EN active-high, 8 ms FAULT deglitch, ILIM tied to IN = 75 mA typ (too low).
R5 VERIFY: pick from datasheet Eq.1 / Fig.22 for IOS(min) > 130 mA (1% part).
Shunt 0.39R x INA180A2 (50 V/V) = 19.5 mV/mA -> 100 mA = 1.95 V at P0.3.
LNA_VCC continues to the LNA bias filter on the RF sheet.

D Lamp rail



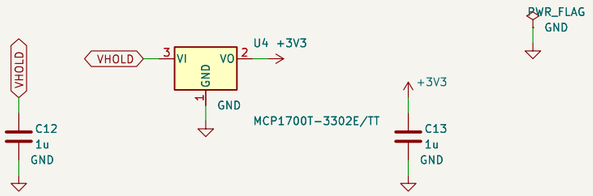
Lamps hang on VMAIN (not the hold-up): they go dark in every OOK gap
and in a dropout instead of draining the supercap. 48 x ~1 mA max.

E Hold-up: fast 330 uF + 0.1 F supercap, diode-OR



Fast path R9/D3 boots the MCU in ~30 ms. Slow path R10/D4 charges the supercap
(<=18 mA, tau ~22 s). D5 lets the supercap carry VHOLD through a dropout (~4 min @ 0.6 mA).
SCAP_ADC = VSCAP / 2 -> P0.2 (HOLD lamp).

F 3V3 for MCU, switches, INA180, 595s



MCP1700: 1.6 uA Iq, 178 mV dropout. Needs >= 1 uF ceramic on the output.

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Coax DC tap, ideal diode, LNA switch + current monitor, hold-up, 3V3

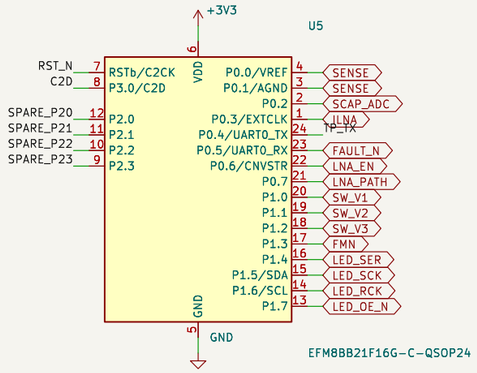
z2labs

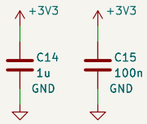
Sheet: /Power/
File: power.kicad_sch

Title: Power

Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/3

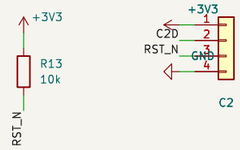
A EFM8BB21F16G-C-QSOP24





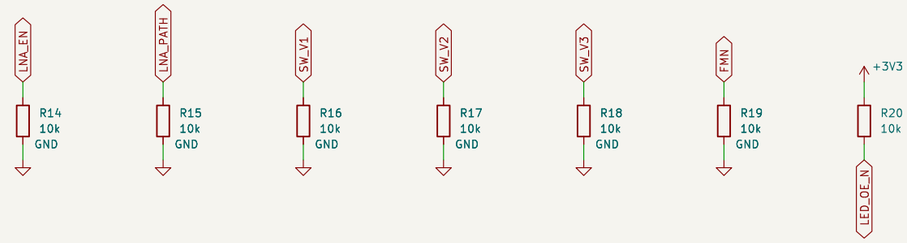
Pin map = firmware/src/hal_efm8bb2.c. P0.0 and P0.1 share SENSE (digital + ADC).
SYSCLK HFOSC0/8 = 3.06 MHz; Snooze when idle (no clock -> no spurs). Keep under the can.

B C2 debug / programming



4-pin 2.54 mm: 3V3, C2D, C2CK(RSTb), GND. Works with a SiLabs debug adapter
or an Arduino-based C2 programmer. Do not add a cap on RSTb (C2 clock).

C Reset-safe defaults (pins are weak-pull-up / Hi-Z until firmware runs)



Pull-downs: LNA off, bypass, branch 0 = THRU, FM notch out -> the box is a plain cable during reset.
OE_N pull-up keeps all lamps dark until the first panel write. FAULT_N pull-up is on the power sheet.

D Test points



TP_TX: firmware drives it low; route UART0 TX here for debug prints if ever needed.

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EFM8BB21F16G, C2 debug, reset-safe pull resistors

z2labs

Sheet: /MCU/
File: mcu.kicad_sch

Title: MCU

Size: A3	Date: 2026-10-26	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/3